

# Simple Blog Application - Project Execution README

## Overview

This project is a full-stack web application that enables users to Create, Read, Update, and Delete (CRUD) blog posts. It uses a modern JavaScript stack (React, Node.js, Express.js, and MongoDB Atlas) with containerized deployment via Docker and cloud hosting on Render.

## Tech Stack

- Frontend: React.js, HTML5, CSS3
- Backend: Node.js, Express.js
- Database: MongoDB Atlas (cloud-hosted NoSQL) with Mongoose
- Deployment/Containerization: Docker, Render
- Version Control: GitHub

## Features

- Create, view, update, and delete blog posts.
- Responsive design with dynamic rendering using React.
- RESTful API communication between frontend and backend.
- Cloud database connectivity with MongoDB Atlas.
- Secure configuration using .env file.
- Containerized deployment using Docker.
- Deployed on Render for public access.

## Project Setup and Execution

1. Clone the repository:

```
git clone https://github.com/M-612/blog-.git
```

2. Backend Setup:

```
cd backend
```

```
npm install
```

Create a .env file with MONGO\_URI and PORT

```
npm start
```

3. Frontend Setup:

```
cd frontend
```

```
npm install
```

```
npm start
```

Application runs on <http://localhost:3000>

4. Deployment:

Backend and frontend hosted on Render with MongoDB Atlas connection.

## **Future Enhancements**

- Add authentication and user roles.
- Integrate comment and like system.
- Use a rich text editor for blog creation.
- Implement search, filter, and tag features.
- Add analytics dashboard and PWA support.
- Extend scalability with Kubernetes.